

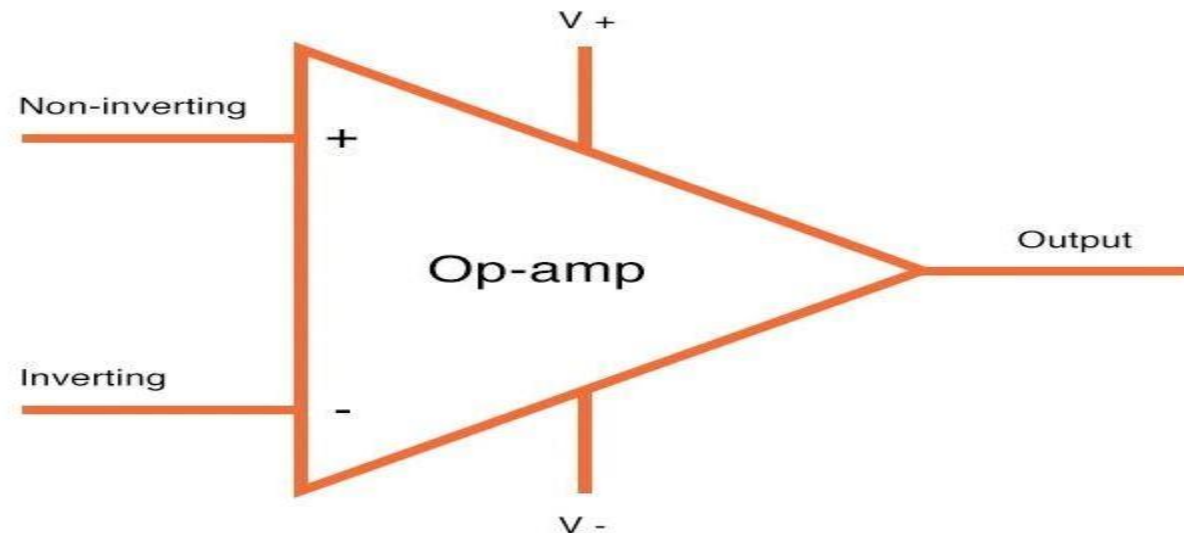
# Chapter 9

Operational Amplifier and Oscillator

# Introduction

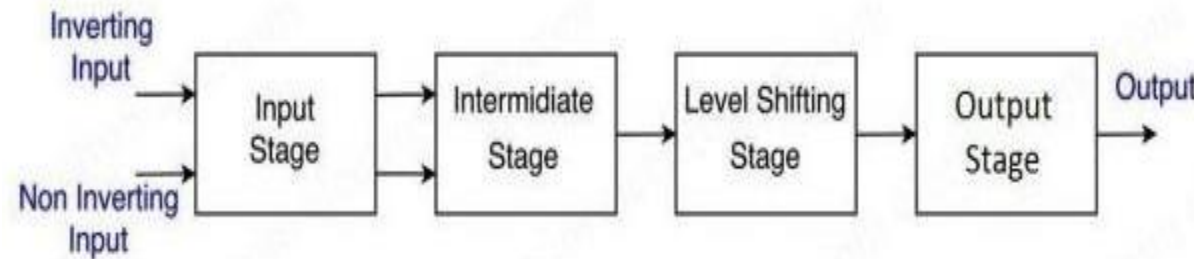
- OP-AMP is basically a multistage amplifier which uses a number of amplifier stages interconnected to each other.
- The integrated op amp offers all the advantage of monolithic integrated circuit such as small size ,high reliability ,reduced cost, less power consumption.

- A voltage amplifying composed of transistors, JFETS, diodes, resistors , and usually one capacitor. Has very high gain, high I/P impedance and low O/P impedance.
- An active circuit element, multistage amplifier can perform mathematical operations such as addition, subtraction, multiplication, division, differentiation and integration etc.
- Real world analog o/p is amplified by OP- amp to increase the signal strength. LM 108, LM 741 etc.



## # Block diagram of op- amp

- The Input Stage is a dual input balanced output differential amplifier.
- Intermediate Stage is a dual input unbalanced output differential amplifier.
- A Level Shifting Stage is used to shift the dc level to zero. where usually an emitter follower with the constant current source is applied.
- The Output Stage is usually a complementary push-pull amplifier.

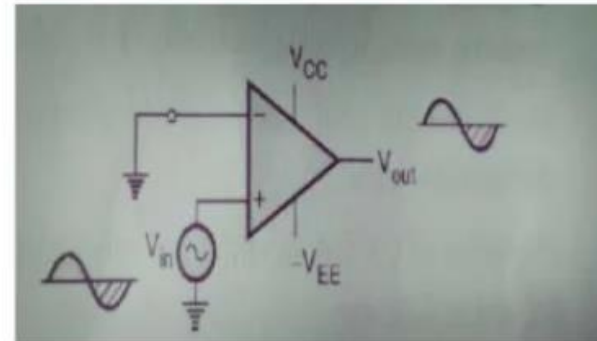
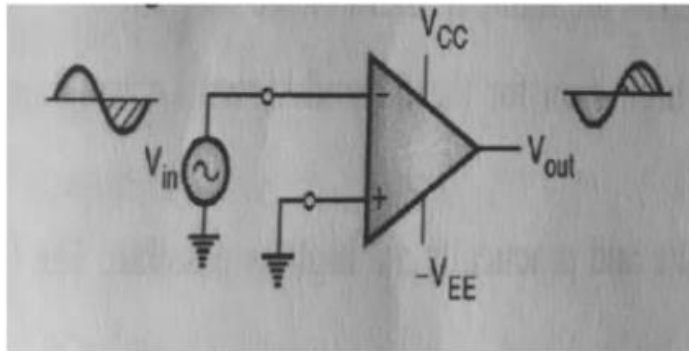


*Opamp Block Diagram*

# OP-AMP input modes

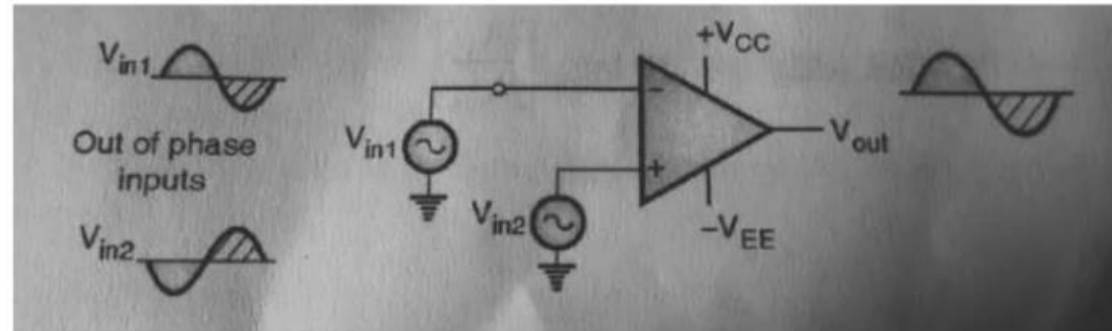
- **Single ended mode**

If the input signal is applied to only one of the inputs and the other input terminal is connected to ground it is said to be operating in single ended mode.



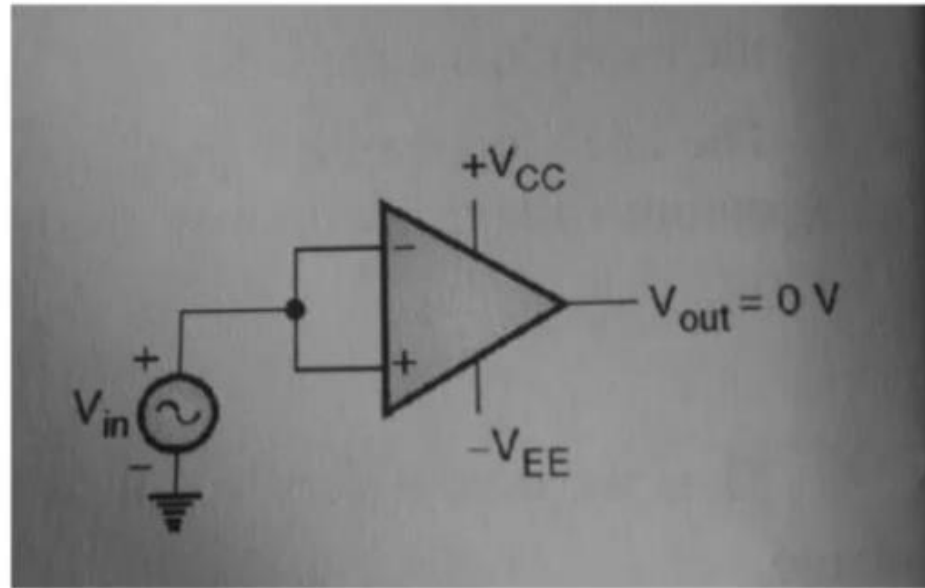
- **Differential mode/double ended**

In differential mode ,two opposite polarity signal are applied to the two inputs of op amp. The difference between the input signal is amplified appears at the output.



- **Common mode**

In the common mode of operation, the same input signal is applied to both the input terminals. Ideally a zero voltage should be produced by the op amp.



# Characteristics of an OP-AMP

- The OP-AMP characteristics(parameters) are important in practice because, we can use them to compare the performance of various op amp ICs and select the best suitable from them for the required application.
- OP-AMP characteristics are classified into two categories namely AC characteristic and DC characteristic .

- **Open loop voltage gain**-It is the differential gain of an OP-AMP in the open loop mode of operation.
- **Input resistance**-It is defined as the equivalent resistance which can be measured at either at inverting or non-inverting terminal with the other terminal connected to ground.
- **Output resistance**-It is the resistance measured by looking into the output terminal of OP-AMP, with the input source short circuited.

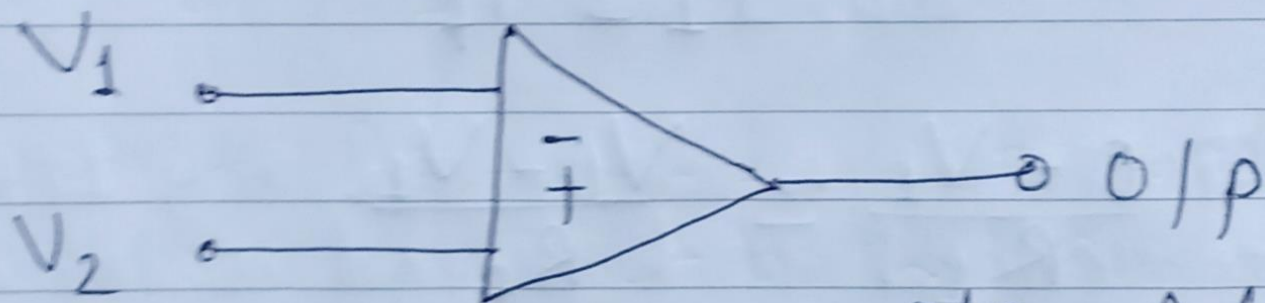
- **Bandwidth**-It is the range over which all signal frequencies are amplified almost equally.
- **Common mode rejection ratio**-It is defined as the ratio of differential gain to common mode gain.
- **Slew rate**-It is defined as the maximum rate of change of output voltage per unit time.
- **Power supply rejection ratio**-It is the change in an OP-AMPs input offset voltage caused by variation in the supply voltage.

- **Input offset voltage**-Ideally, for a zero input voltage output should be zero. But practically it is not so. This is due to unavoidable unbalances inside the OP-AMP.
- **Input bias current**-It is the average of the currents flowing into the two input terminal of the OP-AMP.
- **Input offset current**- It is the algebraic difference between the currents flowing into the inverting and non-inverting terminal of OP-AMP.

# # Characteristics/ Properties of ideal Op-amp

- Date
- 1) Differential or open loop gain (Amplification) is infinity.  
 $A_o = \infty$
  - 2) Input impedance (Resistance) is infinitely high. ( $R_{in} = \infty$ )
  - 3) Output impedance is zero. ( $R_o = 0$ )
  - 4) There is Virtual ground between the input terminals.
  - 5) It has same gain for all frequency range. So, It's frequency bandwidth is infinite.  
 $Bw = \infty$
  - 6) Impedance between any I/P terminal and ground is infinite.
  - 7) Common mode Rejection Ratio (CMRR) is nearly infinite.  
$$CMRR = \frac{\text{Differential gain of op-amp}}{\text{Common mode gain of op-amp}}$$
$$CMRR = \frac{|A_d|}{|A_{cm}|}$$

## # Concept of Virtual Ground.



$$o/p = A(V_2 - V_1)$$

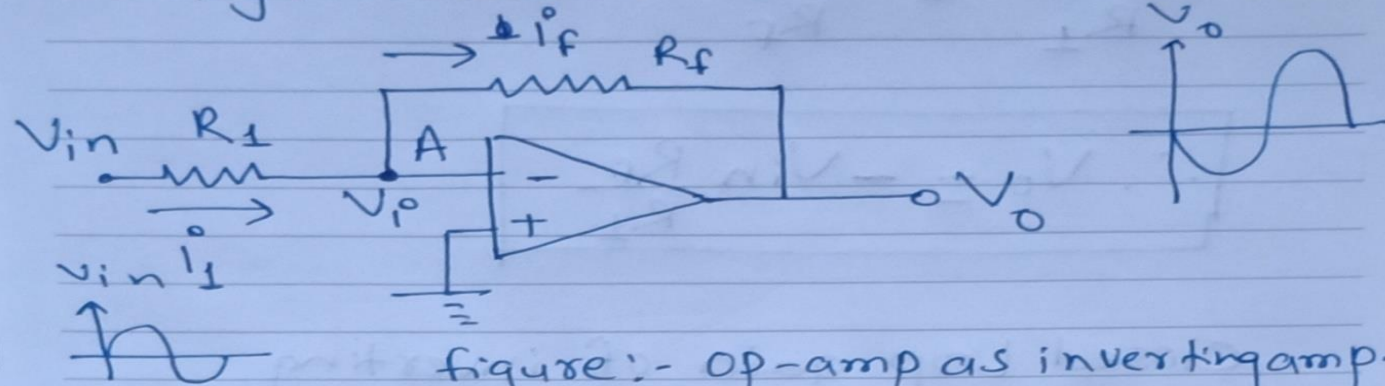
$$\therefore V_0 = A(V_2 - V_1)$$

We know  $A \Rightarrow \infty$ , such that,

$$V_2 - V_1 \approx 0$$

$\therefore V_2 = V_1$  (concept of virtual ground).

# op-amp as inverting amplifier or negative scales.



- I/P is given through  $R_1$  to the inverting terminal.
- feedback resistor  $R_f$  is connected between its o/p and inverting point.

from figure, 
$$i_f = \frac{V_i - V_o}{R_f} \quad \text{--- (1)}$$

$$i_1 = \frac{V_{in} - V_i}{R} \quad \text{--- (2)}$$

Here, point A is at zero voltage due to virtual ground.

$$\therefore V_i = 0 \quad \text{--- (3)}$$

Applying KCL at point A & from (1), (2) & (3)

$$i_1 = i_f$$

Date

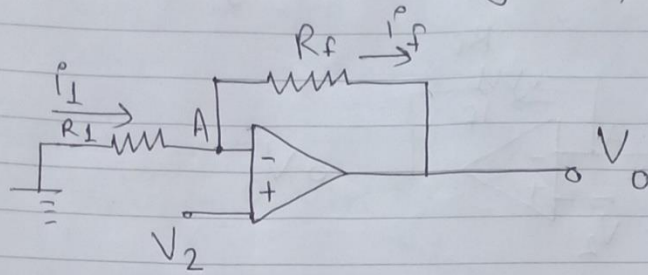
$$\frac{V_{in}}{R_1} = -\frac{V_o}{R_f}$$

$$\therefore V_o = -V_{in} \frac{R_f}{R_1}$$

Closed loop gain of inverting amplifier depends on value of  $R_f$  &  $R_1$ .

$$\left[ \therefore \frac{R_f}{R_1} = \text{gain} \right]$$

Op-amp as Non Inverting<sup>Date</sup> amplifier.



Here point A is Virtually grounded.

$$I_1 = \frac{0 - V_2}{R_1} = -\frac{V_2}{R_1}$$

$$I_f = \frac{V_2 - V_0}{R_f}$$

Applying KCL at A.

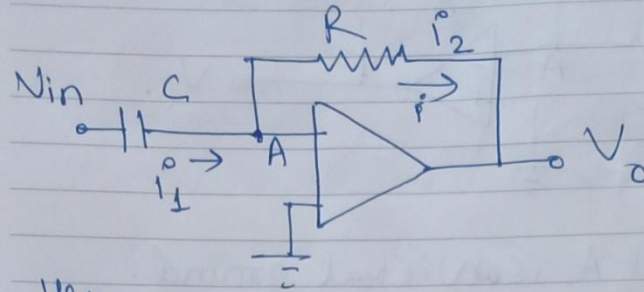
$$I_1 = I_f$$

$$-\frac{V_2}{R_1} = \frac{V_2 - V_0}{R_f}$$

$$V_0 = V_2 \left( 1 + \frac{R_f}{R_1} \right)$$

Non Inverting op-amp with gain =  
 $\left( 1 + \frac{R_f}{R_1} \right) //$

## OP-amp as Differentiator Circuit.



Here,

point A is virtually grounded.

$$\therefore i_1 = C \frac{dV_{in}}{dt}$$

$$i_2 = \frac{0 - V_o}{R} = -\frac{V_o}{R}$$

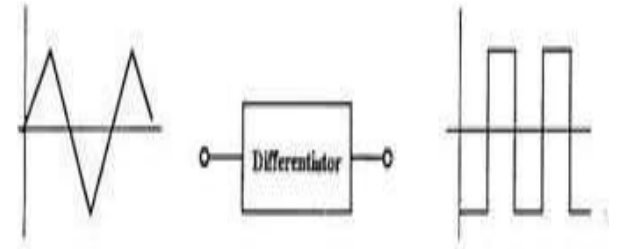
Now, applying KCL at point A

$$C \frac{dV_{in}}{dt} = -\frac{V_o}{R}$$

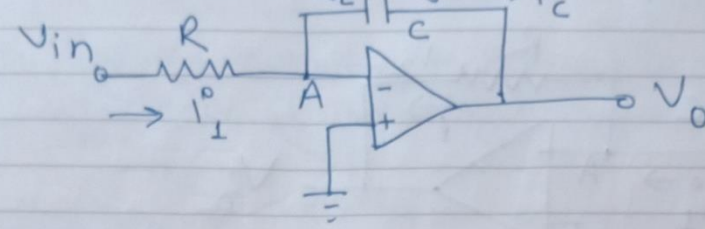
$$\therefore V_o = -RC \frac{dV_{in}}{dt}$$

O/P voltage is proportional to rate of change of I/P voltage.

RC is called time constant or differentiation constant.



# OP-amp as Integrator Date



Here Point A is Virtual ground.

$$\therefore I_1 = \frac{V_{in}}{R} \quad \text{--- (1)}$$

$$I_c = C \frac{dV_c}{dt} \quad \text{--- (2)}$$

Here, Voltage across capacitor  $V_c = 0 - V_o = -V_o$  --- (3)

From (1), (2) & (3) applying KCL

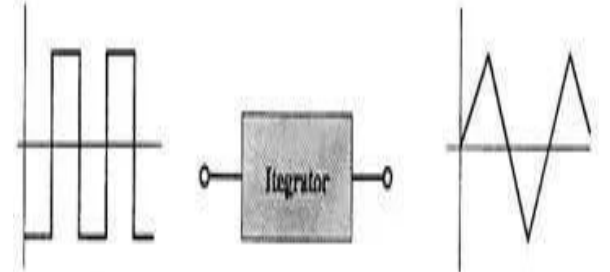
$$I_1 = I_c$$

$$\frac{V_{in}}{R} = -C \frac{dV_o}{dt}$$

$$\frac{dV_o}{dt} = -\frac{V_{in}}{RC}$$

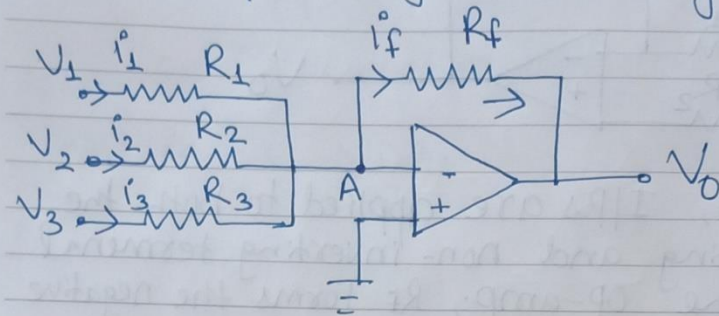
$$\therefore V_o = -\frac{1}{RC} \int V_{in} \cdot dt + A.$$

Here, o/p Voltage is integration of I/p.



## # OP-amp as summing Amplifier/ Summer.

→ Summer circuit provides an output voltage proportional to the algebraic sum of two or more inputs multiplied by a constant gain factor.



Here, point A is Virtually grounded.

$$i_1 = \frac{V_1}{R_1}, \quad i_2 = \frac{V_2}{R_2}, \quad i_3 = \frac{V_3}{R_3}$$

$$\& i_f = \frac{0 - V_o}{R_f} = -\frac{V_o}{R_f}$$

Applying KCL at point A.

$$i_1 + i_2 + i_3 = i_f$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_o}{R_f}$$

$$\therefore V_o = - \left( \frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right)$$

If  $R_1 = R_2 = R_3 = R$ .

$$\therefore V_o = -\frac{R_f}{R} (V_1 + V_2 + V_3) //$$

# Design the amplifier circuit using OP-amp.

$$V_0 = 3V_1 + 2V_2 + V_3$$

We have, For three I/p summing amplifiers.

$$V_0 = - \left( \frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right)$$

[where  $V_1, V_2$  &  $V_3$  are I/ps]

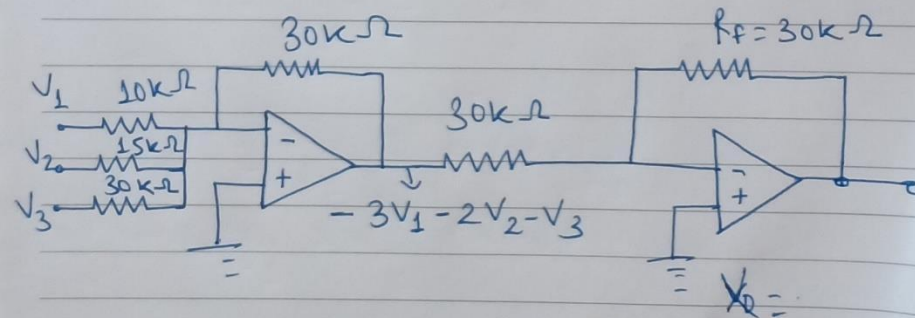
Let,  $R_f = 30k\Omega$

We have,

$$\frac{R_f}{R_1} = 3 \quad \therefore R_1 = 10k\Omega$$

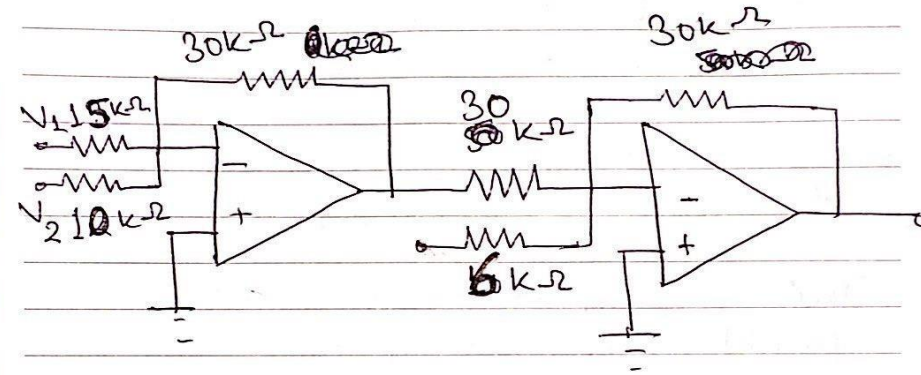
$$\frac{R_f}{R_2} = 2 \quad \therefore R_2 = 15k\Omega$$

$$\frac{R_f}{R_3} = 1 \quad \therefore R_3 = 30k\Omega$$



$$V_0 = 3V_1 + 2V_2 + V_3$$

$$\# \quad V_0 = 2V_1 + 3V_2 - 5V_3$$



let,  $R_f = 30\text{k}\Omega$ ,

$$\therefore R_1 = \frac{30}{2} = 15\text{k}\Omega$$

$$R_2 = \frac{30}{3} = 10\text{k}\Omega$$

$$R_3 = \frac{30}{5} = 6\text{k}\Omega$$

**3. Slew Rate.** Gain bandwidth (GBW) and rise time are both characteristics of small signals, where peak output voltage  $V_{peak}$  is less than 1 V. For many front-end applications this may be adequate, but most power or driver stages must output larger signals. For larger signal outputs,  $V_{peak} > 1$  V, the op-amp's speed is limited by the slew rate.

The slew rate of an op-amp is defined as the maximum rate at which the output voltage can change, no matter how large an input signal is applied.

$$i.e. SR = \left. \frac{dV_{out}}{dt} \right|_{max} \quad \dots(13.26)$$

This is usually measured in V/ $\mu$ s. As illustrated in Fig. 13.23, the instantaneous change in input voltage does not appear instantaneously but slews so many volts in a given period of time.

The slew rate provides a parameter that specifies the maximum rate of change of the output voltage when driven by a large step input signal. If one tried to drive the output at a rate of voltage change greater than slew rate, the output would not be able to change fast enough and would not vary over the full range expected, resulting in signal clipping or distortion. In any case the output would not be an amplified duplicate of the input signal if the op-amp slew rate is exceeded.

Practical op-amps are available with slew rates from 0.1 V/ $\mu$ s to well above 1,000 V/ $\mu$ s. The National semiconductor LH 0063 C has a slew rate of 6,000 V/ $\mu$ s. The slew rate of a 741 C is 0.5 V/ $\mu$ s.

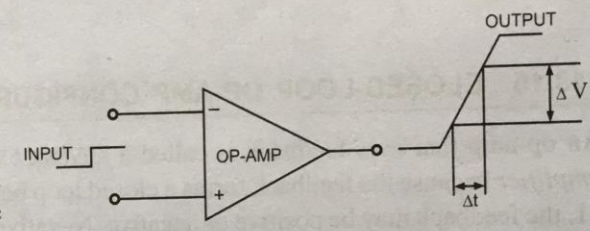


Fig. 13.22

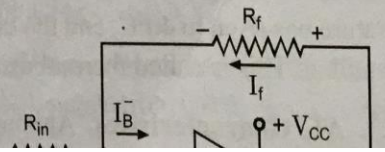
Fig. 13.23

**2. Input Bias Current.** The op-amp's input is a differential amplifier. It may be made of BJTs or FETs. In either case, these transistors are required to be biased, and this takes current. For the BJTs this bias current supplied by the external circuit, is the base bias current, biasing the input transistors into their linear region. The bias current for the FET input op-amp is actually the leakage current across the reverse-biased gate to channel junction.

Manufacturers specify input bias current as the average of the bias current at each input

$$i.e. I_B = \frac{I_{B1} + I_{B2}}{2} \text{ for } V_{out} = 0 \quad \dots(13.22)$$

The value of input bias current  $I_B$  is very small, in the range of a few to a few hundred nanoamperes. It is a dc current and is specified on the op-amp data sheets. For example,  $I_B$  is 500 nA maximum at supply voltages of  $\pm 15$  V dc for the  $\mu$  A741. The value of  $I_B$  has been significantly reduced in later generation op-amps.



**1. Input Offset Voltage.** When the inputs of the op-amp are grounded, there is almost always an *output offset voltage\**, as illustrated in Fig. 13.16, because the input transistors have different  $V_{BE}$  values. For instance, the data sheet of a typical 741C lists a maximum input offset voltage of  $\pm 2$  mV. This difference of 2 mV is an unwanted input signal that is amplified to give an *output offset voltage*. The output offset voltage can be eliminated by applying a voltage of 2 mV to the inverting input as shown in Fig. 13.17. Then the output offset voltage is reduced to zero. Since the offset can have either polarity, it might be necessary to reverse the polarity of the 2 mV.

Thus the input offset voltage  $V_{in(offset)}$  may be defined as that voltage, which is to be applied between the input terminals to balance the amplifier.

The adjustment explained above (Fig. 13.17) could be made to eliminate offset errors if the offset voltages were constant. But these voltages are not constant and depend upon temperature and other parameters such as supply voltage.

Therefore, in most of the practical op-amps extra terminals are provided where an external potentiometer can be connected and adjusted to reduce the offset voltage to zero. For instance, for the 741-type op-amp the manufacturer recommends that a 10 k $\Omega$  potentiometer be placed across offset null pins 1 and 5 and a wiper is connected to the negative supply pin 4, as shown in Fig. 13.18.

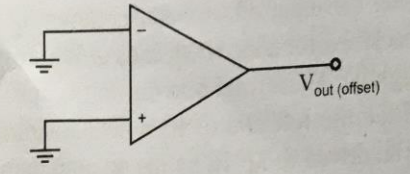


Fig. 13.16 Output Offset Voltage

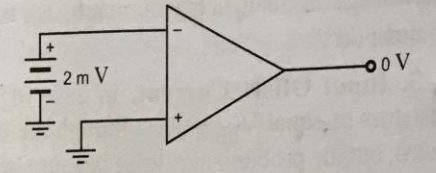


Fig. 13.17 Elimination of Output Offset Voltage

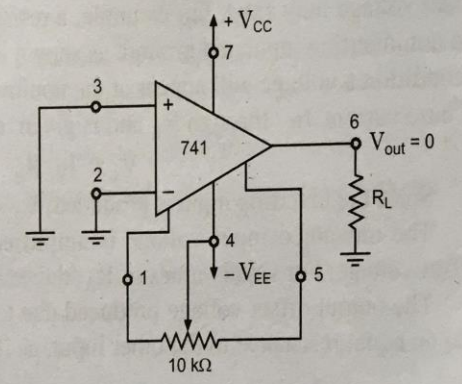


Fig. 13.18 Voltage Offset Null Circuit For The 741

# Oscillators

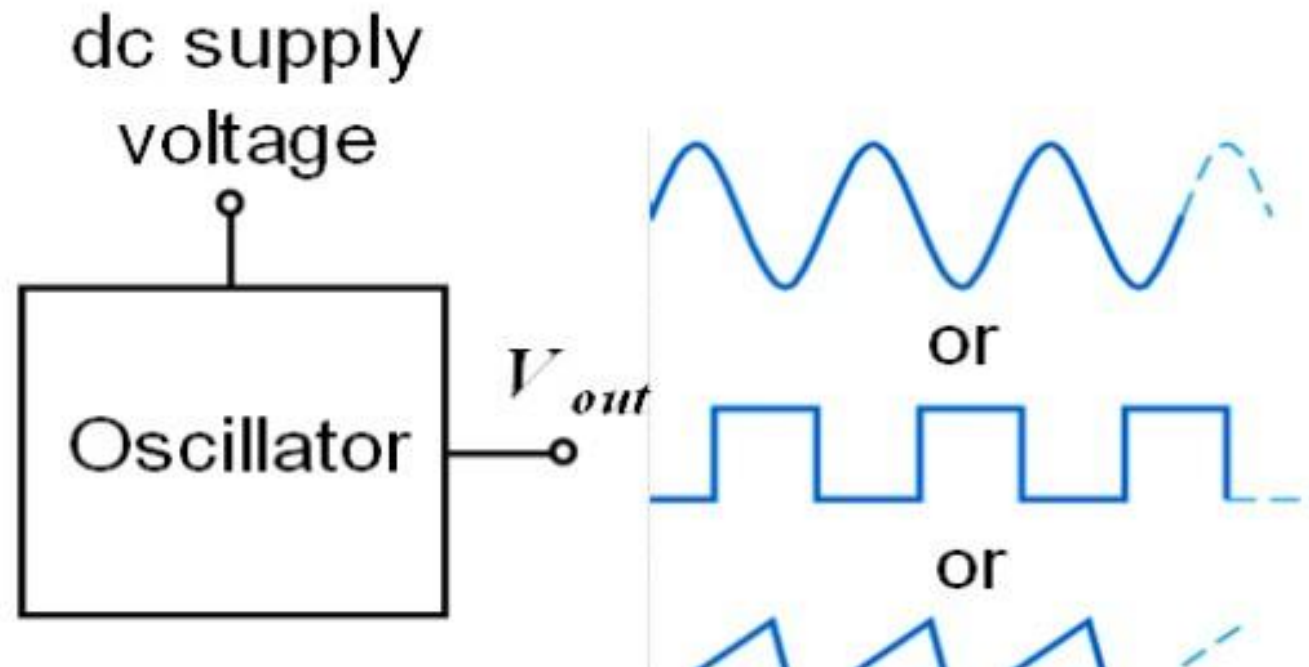
## Introduction

Oscillators are circuits that produce a continuous signal of some type without the need of an input. These signals serve a variety of purposes. Communications systems, digital systems (including computers), and test equipment make use of oscillators.

# Oscillators

Oscillator is an electronic circuit that generates a periodic waveform on its output *without an external signal source*. It is used to convert dc to ac.

The waveform can be sine wave, square wave, triangular wave, and sawtooth wave.



# Oscillators

An oscillator is a circuit that produces a repetitive signal from a dc voltage.

The **feedback oscillator** relies on a positive feedback of the output to maintain the oscillations.

The **relaxation oscillator** makes use of an RC timing circuit to generate a nonsinusoidal signal such as square wave.

## **RC oscillators**

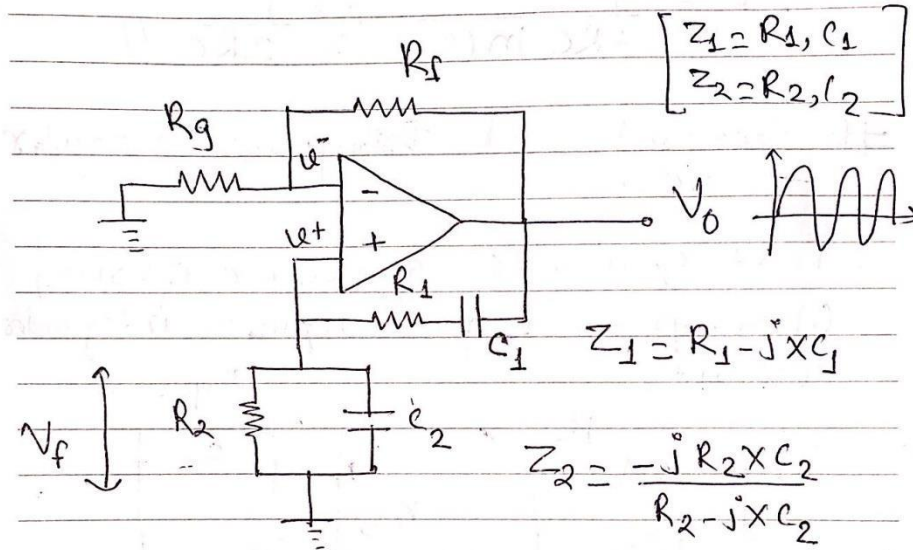
- Wien Bridge
- Phase-Shift

## **LC oscillators**

- Hartley
- Colpitts
- Crystal

## # Wein bridge Oscillator / <sup>Date</sup> Sinusoidal wave form generator

→ Used to generate sinusoidal waveform of frequency and amplitude.



Here, feedback is given to non-inverting  
i/p.  $Z_1$  and  $Z_2$  form  
voltage divider, the feedback  
ratio,  $\beta = \frac{V^+}{V_o} = \frac{Z_2}{Z_1 + Z_2}$

Resistor  $R_g$  &  $R_f$  determine the  
amplifier gain

Now,

Date

$$\beta = \frac{Z_2}{Z_1 + Z_2} = \frac{-jR_2XC_2 / R_2 - jXC_2}{(R_1 - jXC_1) + \left( \frac{-jR_2XC_2}{R_2 - jXC_2} \right)}$$

$$\frac{V^+}{V_0} = \frac{-jR_2XC_2}{R_1R_2 - jXC_1R_2 - jXC_2R_1 - XC_1XC_2 - jR_2XC_2}$$

$$= \frac{R_2XC_2}{jR_1R_2 - jXC_1XC_2 + XC_1R_2 + XC_2R_1 + R_2XC_2}$$

[multiply & divide by j]

$$= \frac{R_2XC_2}{j(R_1R_2 - XC_1XC_2) + (XC_1R_2 + XC_2R_1 + R_2XC_2)}$$

To have  $V_f$  &  $V_0$  ( $V^+$  &  $V_0$ ) in same phase,  $\beta$  must be any real number, which requires imaginary part must be zero.

$$R_1R_2 - XC_1XC_2 = 0$$

$$R_1R_2 = \frac{1}{\omega C_1} \times \frac{1}{\omega C_2}$$

$$\omega = \frac{1}{\sqrt{R_1R_2C_1C_2}}$$

Date

If  $R_1 = R_2 = R$ ,  $C_1 = C_2 = C$  the frequency of oscillation ~~beco~~ become,  $\omega = \frac{1}{RC}$   $\therefore f = \frac{1}{2\pi RC}$

At this condition,

$$X_{C_1} = X_{C_2} = \frac{1}{\omega C} = \frac{1}{\frac{1}{RC} \times C} = R$$

Substituting  $R_1 = R_2 = X_{C_1} = X_{C_2} = R$

$$\beta = \frac{V_f}{V_o} = \frac{R^2}{R^2 + R^2 + R^2 + j(R^2 - R^2)}$$

$$\therefore \beta = \frac{R^2}{3R^2} = \frac{1}{3}$$

$\therefore$  The amplifiers should provide a gain of 3 to make the value of  $A\beta$  or loop gain equal to unity.

Since,  $A = 3$

We require,  $1 + \frac{R_f}{R_g} = 3$

$$\therefore \frac{R_f}{R_g} = 2 //$$

## # RC Phase Shift Oscillator using op-amp.

RC oscillator  $\rightarrow$  Phase shift oscillator

$\rightarrow$  Wien bridge , ,  
(audio frequency oscillators)

$\rightarrow$  Simple circuit, Frequency stability very good.

$\rightarrow$  Low Distortion.

$V_o = \text{o/p Voltage}$

$V_f = \text{i/p} = \text{feedback voltage.}$

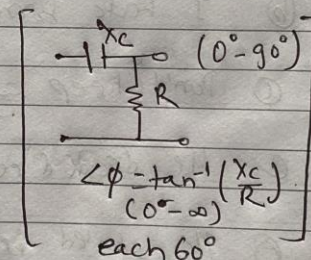
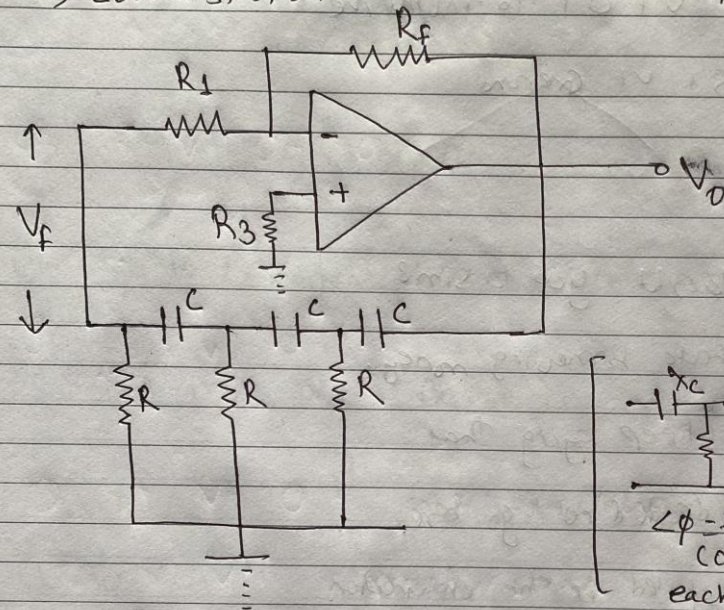


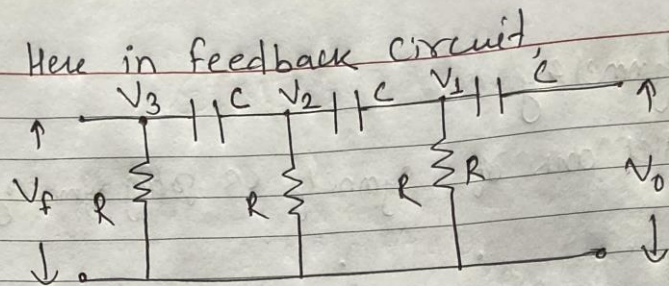
Figure: RC phase shift oscillator

Here, op-amp acts as inverting amplifier, provides  $180^\circ$  phase shift and feedback network RC provides  $180^\circ$  phase shift (each  $60^\circ$ ).

Gain of op-amp is  $A$  and feedback circuit is  $\beta$ .

For oscillation

$$A\beta = 1$$



Applying KCL at node  $V_3$ .

$$\frac{V_3}{R} + \frac{V_3 - V_2}{-jX_C} \quad \left[ \because V_3 = V_f \right]$$

$$\frac{V_f}{R} + \frac{V_f - V_2}{-jX_C} = 0 \quad \text{--- (1)}$$

Applying KCL at node  $V_2$ ,

$$\frac{V_2 - V_1}{-jX_C} + \frac{V_2 - V_f}{-jX_C} + \frac{V_2}{R} = 0 \quad \text{--- (2)}$$

Applying KCL at node  $V_1$ ,

$$\frac{V_1 - V_2}{-jX_C} + \frac{V_1 - V_o}{jX_C} + \frac{V_1}{R} = 0 \quad \text{--- (3)}$$

Solving (1), (2) & (3) we get

$$\frac{V_f}{V_o} = \beta = \frac{R^3}{(R^3 - 5RX_C^2) + j(X_C^3 - 6R^2X_C)}$$

Here,  $\beta$  must be Real,

i.e.  $X_C^3 - 6R^2X_C$  must be zero.

$$X_C^2 = 6R^2$$

$$\Rightarrow \left( \frac{1}{2\pi f C} \right)^2 = 6R^2$$

$$\therefore f = \frac{1}{2\pi RC\sqrt{6}} // \text{--- (4) [Frequency of RC phase Shift Oscillator]}$$

$\therefore$  For Oscillation  $\beta$  must be only real value. (Taking imaginary term as zero).

$$\beta = \frac{R^3}{(R^3 - 5RXC^2)}$$

$$= \frac{R^3}{(R^3 - 5R \times 6R^2)}$$

$$\left[ \begin{array}{l} \text{we have,} \\ XC^2 = 6R^2 \end{array} \right]$$

$$= \frac{R^3}{R^3 - 30R^3}$$

$$= -\frac{R^3}{29R^3} = -\frac{1}{29} //$$

For Oscillation we know,

$$A\beta = 1$$

$$\text{Here } \beta = -\frac{1}{29} \therefore A \Rightarrow -29$$

We have,

$$A = -\frac{R_f}{R} \quad [\text{Inverting Amplifier}]$$

$$\therefore \boxed{R_f = 29R}$$

Required condition for RC phase shift oscillator.

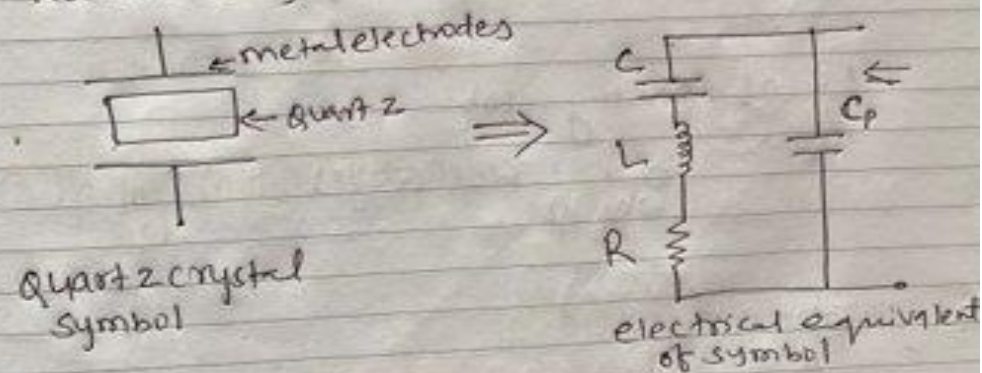
## # Crystal oscillator

In RC oscillator, Frequency may change due to,

- change in temperature
- change in power supply voltage
- change in component values

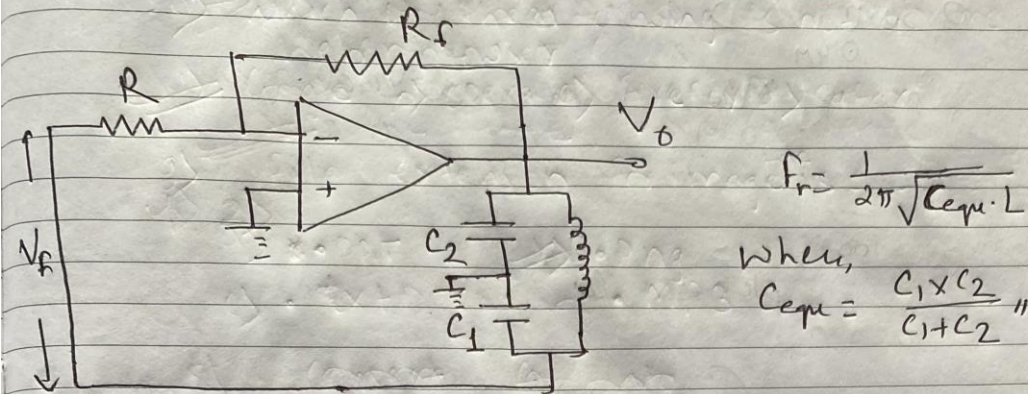
So, for stability and quality factor ( $Q$ ) crystal oscillators are used. ( $Q \rightarrow \infty$ )

- Used in digital circuit, Telecommunication,
- Used in smart phones, microprocessors, microcontrollers for generating clock frequency of different ranges.
- Works on principle of <sup>inverse</sup> piezoelectric effect.
- Rochelle salt, Quartz etc are used.



- Used for High frequency application.
- Hartley oscillator, Colpitts oscillator.

## # Colpitts Oscillator.



$$f_r = \frac{1}{2\pi\sqrt{C_{eq}\cdot L}}$$

where,

$$C_{eq} = \frac{C_1 \times C_2}{C_1 + C_2} //$$

Figure 1 - Colpitts oscillator.

Here, op-amp provides  $180^\circ$  phase shift and

LC tank circuit provides  $180^\circ$ .

$\therefore$  Total phase shift =  $360^\circ$

Here, In feedback circuit  $(\beta) = \frac{V_f}{V_o}$

$$\beta = \frac{I \times C_1}{I \times C_2}$$

$$\beta = \frac{1/2\pi f_r C_1}{1/2\pi f_r C_2}$$

$$\therefore \beta = \frac{C_2}{C_1} //$$

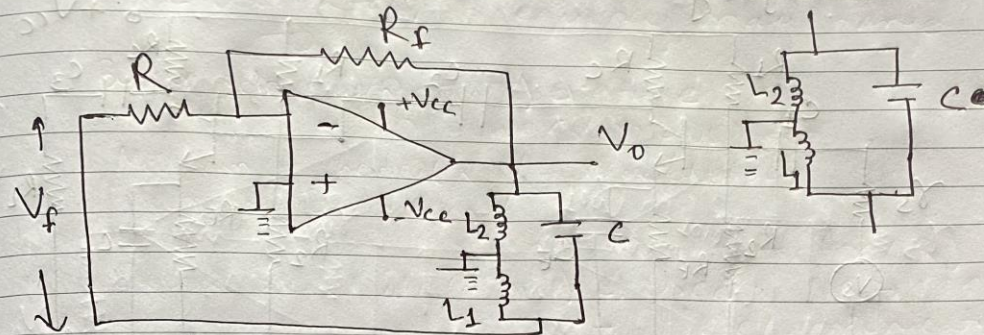
Condition for oscillation,  $A\beta = 1$

$$A = C_1/C_2$$

If  $\beta = -1, \therefore A = -1$

$$\therefore R_f = R //$$

## # Hartely Oscillator



Here, LC circuit provides  $180^\circ$  phase shift, it acts as resonant filters and passes only desired frequency. OP-amp acts as inverting amplifier ( $180^\circ$  phase shift).

Frequency of oscillation,

$$f_r = \frac{1}{2\pi\sqrt{L_{eq}C}}$$

For oscillation total phase shift must be  $0^\circ$  or  $360^\circ$ .

$$\text{Now, } \beta = \frac{V_f}{V_o} = \frac{I \times L_1}{I \times L_2} = \frac{X_{L1}}{X_{L2}}$$

$$\beta = \frac{2\pi f_r L_1}{2\pi f_r L_2} = \frac{L_1}{L_2} \quad \left[ \begin{array}{l} \text{Where} \\ f_r = \text{resonant frequency} \end{array} \right]$$

We have,  $A\beta = 1$

$$A = \frac{L_2}{L_1} = L_2/L_1$$

$$\text{Here, } A = -\frac{R_f}{R} = \frac{L_2}{L_1} \quad \left[ \text{Ratio must be satisfied} \right]$$

If  $A = 1$ ,  $[R_f = R]$ , If  $\beta = -1$ ,

where,  
 $L_{eq} = L_1 + L_2 + 2M$   
 $M = 0$  if not given  
 Mutual inductance.

# Definition of Multivibrator

A multivibrator is an electronic circuit that is used to implement two state systems like oscillators, timers and flip-flops.

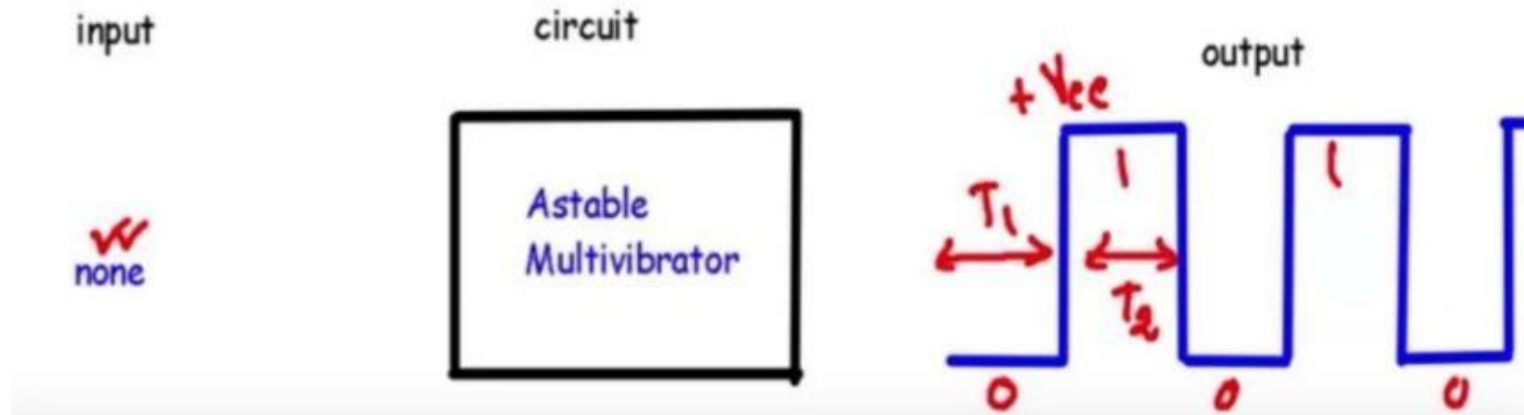
It is characterized by two amplifying devices (transistors or other devices) cross-coupled by resistors or capacitors

# Classification of Multivibrator

- There are three types of multivibrator circuits depending on the circuit operation:
  - – **Astable – neither state is stable.**
  - – **Monostable - one of the states is stable, but the other is not.**
  - – **Bistable – it remains in either state indefinitely.**

# Astable Multivibrator

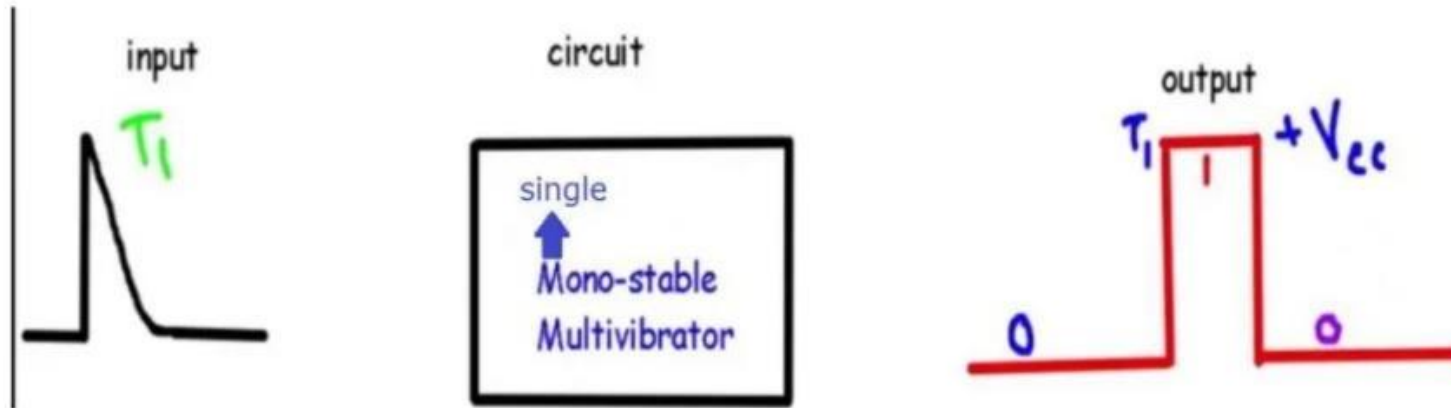
- A Multivibrator that generates square waveform without using external triggering pulse is known as Astable multivibrator. It also known as **Free-running Multivibrator**.



# Monostable Multivibrator

## Definition:

A Multivibrator which has one stable state and one temporary quasi-stable state and level transition depends on external triggering pulse is known as Monostable Multivibrator.



## # Monostable Multivibrator using op-amp.

### Monostable Multivibrator

- Has only one stable state, and whenever the trigger signal is applied then it goes to unstable state.
- If stable state is low state then whenever trigger signal is applied, the output goes to high state & vice versa.

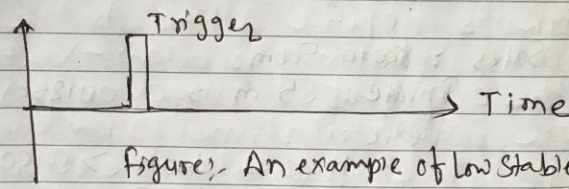
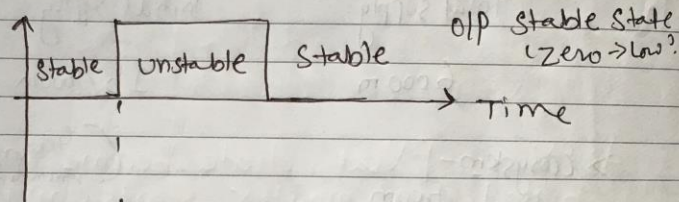


Figure: An example of low stable state

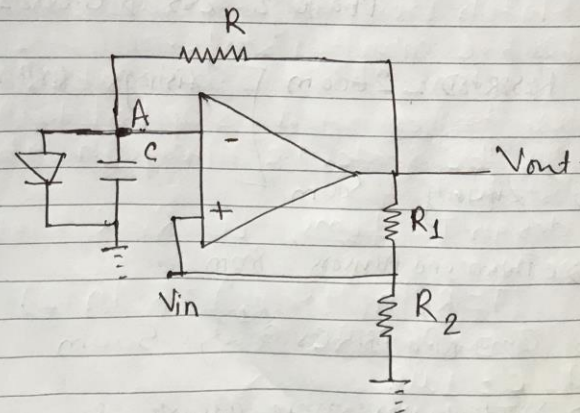
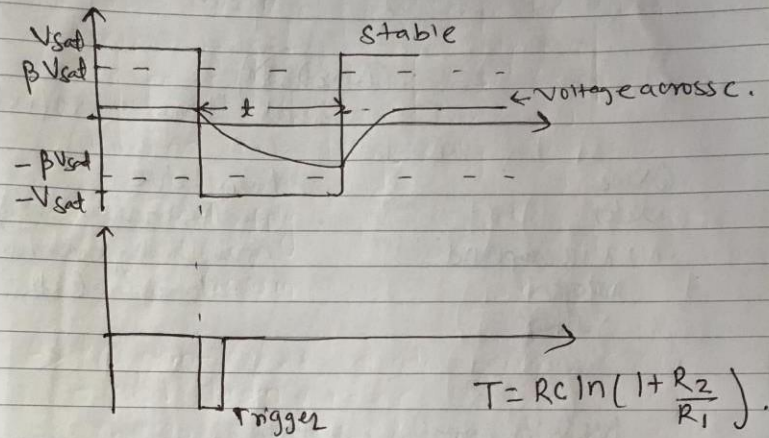


Figure: Monostable Multivibrator

→ Diode is parallel with capacitor, Trigger Signal is applied at  $V_{in} = (V_f) \cdot [V_{in} = V_f]$



Let, O/P voltage is at  $V_{sat}$  (Stable state  $\Rightarrow 1$ )

$$\therefore V_{in} = V_f = \frac{V_o \times R_2}{R_1 + R_2} = \frac{V_{sat} \times R_2}{R_1 + R_2}$$

$$\therefore V_{in} = \beta V_{sat} \quad \left[ \beta = \frac{R_2}{R_1 + R_2} \right]$$

A is Virtually grounded, So, Voltage across diode is positive voltage  $\beta V_{sat}$  which is also appears at capacitor (D11C)

Now,

Trigger signal is applied at node  $V_{in}$ , then output at  $V_{in}$  decreases, when it is less than the voltage at inverting node, output changes its state from  $V_{sat}$  to  $-V_{sat}$  [Here, Trigger voltage is  $-V_e$  signal] at this time voltage at  $V_{in} = -\beta V_{sat}$ .

As  $V_{in} = -\beta V_{sat}$ , diode become reverse biased and capacitor starts charging towards negative voltage. As the voltage across capacitor goes below  $-V_{sat} \times \beta$ , the output of comparator goes to  $+V_{sat}$ . Here for certain time ( $t$ ) the output goes to unstable state. the output will switch from negative saturation to positive saturation state.

For time ' $t$ ' it is in unstable state, Here, trigger signal is applied to make unstable state else always in stable state.

To design  $T = 10\text{ms}$ , Let,  $R_1 = R_2 = 10\text{k}\Omega$   
we have,

$$T = RC \ln(1 + R_2/R_1) = RC \ln(2)$$

$$T = 0.693 RC$$

$$\text{Let, } C = 1\mu\text{f}$$

$$T = 10\text{ms}$$

$$\therefore R \approx 14.4\text{k}\Omega //$$

# Astable Multivibrator (Free or Running

mu

# Generation of Square Wave using an op-amp.

→ Produces Square Wave of desired frequency and amplitude.

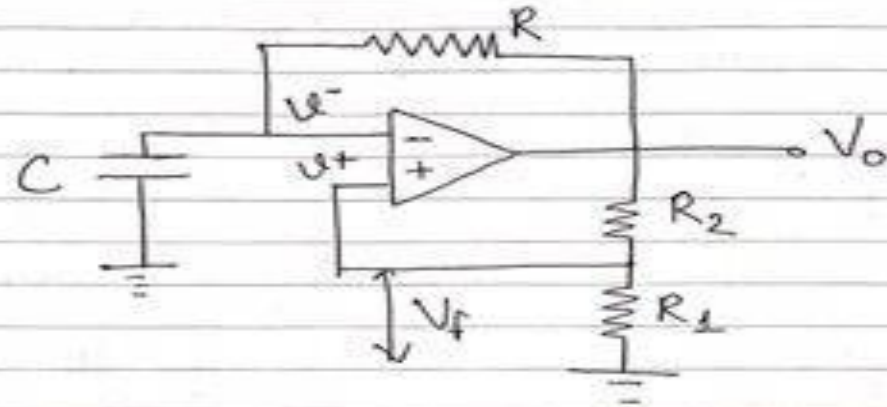


Figure:- Square wave generator using op-amp.

- It consists of op-amp (as comparator), capacitor & resistors.
- Resistor 'R' forms negative feedback loop between I/P & O/P.
- Voltage drop across  $R_1$  is fed to the non-inverting terminal of op-amp to provide positive feedback to fulfill Barkhausen criteria.

$$V_f = \frac{R_1}{R_1 + R_2} \times V_o = \beta V_o \left[ \because \beta = \frac{R_1}{R_1 + R_2} \right]$$

where,  $\beta$  = feedback factor.

→ Operation depends on charging and discharging of capacitor.

Let, O/P voltage of ~~an~~ comparator are equal in magnitude and opposite in polarity  $= \pm V_{max}$ .

1<sup>st</sup> Let, O/P voltage is  $+V_{max}$ , then feedback to non inverting I/P is.

$$V^+ = V_{max} \times \frac{R_1}{R_1 + R_2} = \beta V_{max}.$$

The capacitor begin to charge through  $R$  towards  $V_{max}$ , when capacitor reaches  $+\beta V_{max}$  the comparator switches its state and output is  $-V_{max}$ .

Now, Voltage at feedback (non-inverting I/P) becomes

$$V^+ = \left( \frac{R_1}{R_1 + R_2} \right) \times (-V_{max})$$
$$= -\beta V_{max}.$$

During this capacitor discharged fully. After this  $-V_{max}$  is fed to noninverting I/P and capacitor gets negatively

charged upto  $-\beta V_{max}$ . When capacitor reaches to  $-\beta V_{max}$ , output switched to  $+V_{max}$ .

The process goes on (repeats) continuously with a resulting square wave of amplitude  $\pm V_{max}$ .

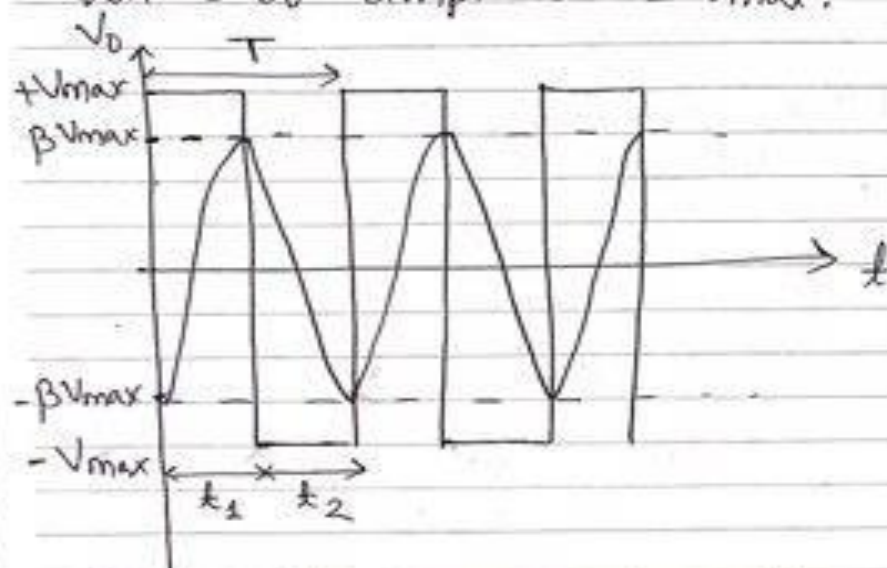


Figure 1 - Square wave generation (charging & discharging of capacitor).

For determination of Time period & frequency,  $T = t_1 + t_2$

For charging of capacitor,

$$V_c(t_1) = V_{applied} - [V_{applied} - V_c(\text{initial})]e^{-t/RC}$$

— (1) [at  $t = t_1$ ]

Where,

Date

$$V_c(t_1) = \beta V_{max}$$

$$V_c(\text{initial}) = -\beta V_{max}$$

$$V_{\text{applied}} = V_{max}$$

Now, (1) becomes,

$$\beta V_{max} = V_{max} - [V_{max} - (-\beta V_{max})] e^{-t_1/RC}$$

$$\text{or } \beta V_{max} = V_{max} [1 - e^{-t_1/RC}]$$

$$\text{or } V_{max}(1+\beta) e^{-t_1/RC} = V_{max}(1-\beta)$$

$$\text{or } e^{-t_1/RC} = \frac{1-\beta}{1+\beta}$$

$$\text{or } t_1/RC = \ln\left(\frac{1+\beta}{1-\beta}\right)$$

$$t_1 = RC \ln\left(\frac{1+\beta}{1-\beta}\right)$$

$$\text{Total time period } T = 2t_1 \quad [t_1 = t_2]$$

$$T = 2RC \ln\left(\frac{1+\beta}{1-\beta}\right)$$

$$\therefore f = \frac{1}{2RC \ln(1+\beta/1-\beta)}$$

$$\text{If } R_1 = R_2 = R, \beta = \frac{1}{2}$$

$$\therefore T = 2RC \ln(3) \quad \text{[scribbled out]$$

$$\therefore F = \frac{1}{2RC \ln(3)} \approx \frac{1}{2RC} //$$

# Generation of ~~Triangular~~ Triangular Wave.

→ First generate square wave using op-amp and give output to integrator circuit.

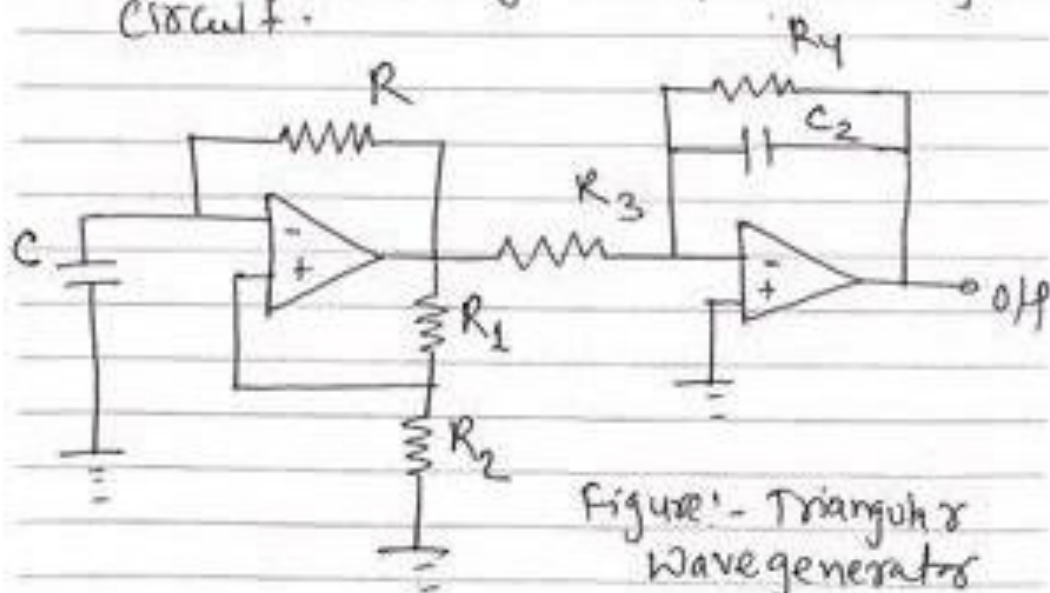


Figure 1 - Triangular Wave generator

→ Frequency is same as square wave.

**Example 11.17. Design the R-C elements of a Wien bridge oscillator for operation at  $f_0 = 10$  kHz.**

*[M.D. Univ. Analog Electronic Circuits, December-2006]*

**Solution:** Using values of  $R_1$  equal to  $R_2$  and  $C_1$  equal to  $C_2$  and taking each of  $R_1$  and  $R_2$  of 200 k $\Omega$ ,

$$\text{Required value of } C = \frac{1}{2\pi f R} = \frac{1}{2\pi \times 10 \times 10^3 \times 200 \times 10^3} = 79.6 \text{ pF Ans. Refer to Eq. (11.33)}$$

We can use  $R_3 = 400$  k $\Omega$  and  $R_4 = 200$  k $\Omega$  **Ans.**

**Example 11.18. Find the frequency of oscillations of a Wien bridge oscillator with  $R = 20$  k $\Omega$  and  $C = 1,000$  pF.**

**Solution:**

$$R = R_1 = R_2 = 20 \text{ k}\Omega = 20,000 \Omega$$

$$C = C_1 = C_2 = 1,000 \text{ pF} = 1 \times 10^{-9} \text{ F}$$

$$\text{Frequency of oscillation, } f = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 20,000 \times 1 \times 10^{-9}} = 7.96 \text{ kHz Ans.}$$

**Example 11.16.** The parameters of a crystal oscillator equivalent circuit are  $L_s = 0.8 \text{ H}$ ;  $C_s = 0.08 \text{ pF}$ ,  $R_s = 5 \text{ k}\Omega$  and  $C_p = 1.0 \text{ pF}$ . Determine the resonance frequencies  $f_s$  and  $f_p$ .

[U.P.S.C. I.E.S. Electrical Engineering II, 1999; Pb. Technical Univ. Analog Electronics, May-2007; Rajasthan Univ. 2003, 2005, 2006]

**Solution:**

Series resistance,  $R_s = 5 \text{ k}\Omega = 5,000 \Omega$

Series inductance,  $L_s = 0.8 \text{ H}$

Series capacitance,  $C_s = 0.08 \text{ pF} = 8 \times 10^{-14} \text{ F}$

Parallel capacitance,  $C_p$  or  $C_M = 1.0 \text{ pF} = 1 \times 10^{-12} \text{ F}$

$$\text{Series resonant frequency, } f_s = \frac{1}{2\pi\sqrt{L_s C_s}} = \frac{1}{2\pi\sqrt{0.8 \times 8 \times 10^{-14}}} = 629 \text{ kHz Ans.}$$

$$\text{Parallel resonant frequency, } f_p = \frac{1}{2\pi} \sqrt{\frac{1 + C_s/C_p}{L_s C_s}} = \frac{1}{2\pi} \sqrt{\frac{1 + 8 \times 10^{-14}/1 \times 10^{-12}}{0.8 \times 8 \times 10^{-14}}} = 654 \text{ kHz Ans.}$$

4b) Ans

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$$h_{fe} = 100$$

$$h_{ie} = 1.175 k\Omega$$

$$h_{oe} = 20 \mu A/V$$

$$A_i = ? \quad A_v = ? \quad Z_o = ? \quad Z_{in} = ?$$

$$1) \quad Z_{in} = R_B \parallel h_{ie} \approx h_{ie} = 1.175 k\Omega \quad \text{or } (1.17 k\Omega) \\ \text{or do } (R_B \parallel h_{ie}) = (330 k\Omega \parallel 1.175 k\Omega)$$

$$2) \quad r_e = \frac{1}{h_{oe}} = \frac{1}{20 \mu A/V} = 50 k\Omega$$

$$Z_o = \frac{1}{h_{oe}} \parallel R_e = (50 k\Omega \parallel 2.7 k\Omega) = 2.56 k\Omega \\ \text{or } Z_o \approx R_e = 2.7 k\Omega \approx 2.5 k\Omega$$

$$3) \quad A_i \approx h_{fe} = 100$$

$$4) \quad A_v = \frac{-h_{fe} (R_C \parallel \frac{1}{h_{oe}})}{h_{ie}}$$

$$= \frac{-100 (2.56 k\Omega)}{1.171 k\Omega} = -218.61$$

5>a)

Given, For class B amplifier,

$$\text{Peak voltage } (V_m) = 10 \text{ V}$$

$$\text{Power supply } (V_{cc}) = 20 \text{ V}$$

$$\text{Load } (R_L) = 20 \Omega$$

$$\text{efficiency } (\eta) = ?$$

$$\text{I/P power } (P_{in}) = ?$$

$$\text{O/P power } (P_{out}) = ?$$

We have, ①

$$\eta = \frac{V_m}{V_{cc}} \times 78.5\% \quad (\text{for class B})$$

$$= \frac{10}{20} \times 78.5\%$$

$$\eta = 39.25\% //$$

[Note,  
To derive  $\eta$ , you  
can use ① or ②.]

We have,

$$(P_s) \text{ I/P power or, supply power } (P_{in}) = \frac{2 \times V_m \times V_{cc}}{\pi \times R_L}$$

$$P_{in} = \frac{2 \times 10 \times 20}{\pi \times 20} = 6.369 \text{ W} //$$

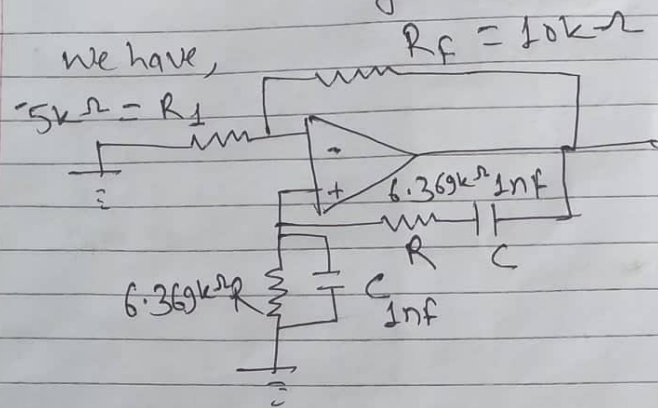
$$\text{O/P power } (P_{out}) \text{ or Load power } (P_L) = \frac{\left(\frac{V_m}{\sqrt{2}}\right)^2}{R_L}$$

$$\therefore P_{out} = \frac{V_m^2}{2 R_L} = \frac{10^2}{2 \times 20} = 2.5 \text{ W} //$$

$$\therefore \eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{2.5}{6.369} \times 100 = 39.25\%$$

Answer

6) 6) given  
 $f = 25 \text{ kHz}$



$$R = \frac{1}{2\pi f C}, \quad \text{Let, } C = 1 \text{ nF}$$

$$\therefore R = \frac{1}{2 \times \pi \times 25 \times 10^3 \times 1 \times 10^{-9}} = 6369.42 \Omega$$

$$\therefore R = 6.369 \text{ k}\Omega //$$

we know for Weinbridge oscillator,

$$\text{Gain } A = \left(1 + \frac{R_f}{R_1}\right)$$

Again we know,

$$3 = 1 + \frac{R_f}{R_1} \quad \therefore \frac{R_f}{R_1} = 2$$

$$\therefore R_f = 2R_1 \quad \text{or, } R_1 = \frac{R_f}{2}$$

$$\text{Let, } R_f = 10 \text{ k}\Omega, \therefore R_1 = 5 \text{ k}\Omega //$$

Note, you can let any parameter, normally  $R_f$  is taken.

